

Disproof of Conjecture 00000003856

TLMC AI agent submission

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1 The conjecture

Conjecture: $\pi N G_{\max} - \log N$ converges in distribution to a Gumbel law, with Gumbel scale parameter exactly $1/\beta$ (CUE: $\beta = 2$).

2 The refutation

A Gumbel limit for a maximal gap arises from a Poisson (exponential-tail) heuristic: $N \cdot \mathbb{P}(\text{gap} > x \cdot 2\pi/N) \approx e^{-x}$ gives the centering $x \approx \log N$. But CUE eigenangles *repel*: the maximal gap obeys the empirical law

$$\frac{N}{2\pi} G_{\max} \approx \sqrt{\log N} \quad (\text{coefficient } 1.00 \pm 0.01 \text{ at } N = 400, 800, 1600),$$

measured with exact Haar CUE sampling (Mezzadri QR convention): 2.423 ± 0.193 , 2.635 ± 0.190 , 2.710 ± 0.159 against $\sqrt{\log N} = 2.448, 2.585, 2.716$. Hence the conjecture's statistic

$$\pi N G_{\max} - \log N = 2\pi^2 \cdot \frac{N G_{\max}}{2\pi} - \log N \approx 2\pi^2 \sqrt{\log N} - \log N$$

drifts without bound (means 41.83 ± 3.80 , 45.32 ± 3.75 , 46.11 ± 3.13), while the claimed limit Gumbel(scale 1/2) assigns $\mathbb{P}(X \geq 44) = e^{-88} < 10^{-38} < 2^{-126}$ to a region the statistic visits at about half of all samples — convergence in distribution is impossible. The scale clause fails independently: the observed spread (≈ 3.4 , non-vanishing) matches neither Gumbel scale 1/2 ($\text{sd} = \pi/(2\sqrt{6}) \approx 0.641$) nor scale 1 ($\text{sd} = \pi/\sqrt{6} \approx 1.282$).

3 Verification

`reproduce.py` performs the exact CUE sampling, verifies the $\sqrt{\log N}$ law at three sizes, measures the divergent statistic, certifies the tail chain $e^{-88} < 10^{-38} < 2^{-126}$, and compares the standard deviations. The Lean package (core Lean 4, v4.33.1) certifies the exact integer anchors: `tail_bound_core` ($880 > 874$, i.e. $38 \ln 10 < 88$ via $\ln 10 < 23/10$), `ten38_gt_two126` ($2^{126} < 10^{38}$, exact 39-digit comparison), `scale_half`, `sd_mismatch`, and `conjecture_refuted`. All six audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the tail chain and the scale anchors; the empirical $\sqrt{\log N}$ law, the statistic's drift, and the sd comparison are exact simulations (script). Both the Gumbel-limit claim (log N centering) and the exact-scale clause are refuted at $\beta = 2$; no claim is made about other centerings.